ADS401 HCI EVALUATION · GROUP H

# Usability Evaluation of the XJTLU Museum AI System

A broader usability evaluation, including a 30-student post-visit experience experiment.

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kinger-ai.github.io

| SAMPLE | GROUPS | OUTCOMES | ADJUSTED RESULT |
|---|---|---|---|
| **30** | **15 + 15** | **9 items** | **0 sig.** |
| XJTLU students, 17 undergraduate and 13 postgraduate. | AI-guide and no-AI comparison groups. | Q7–Q15 ratings from 1 to 5. | No Holm-adjusted p-value was below .05. |

## 01 PROJECT CONTEXT

**Overall topic:** How usable is the museum AI guide, and how does AI-supported guidance shape the museum visit?

How do post-visit experience ratings differ between students who used the AI guide and those who did not?

This 30-person questionnaire is one embedded experiment, not the entire project.

## 02 DESIGN

1. Consent and participant information.
2. AI-guide or no-AI museum condition.
3. Visit the same exhibition context.
4. Complete the anonymous post-visit survey.
5. Compare Q7–Q15 ratings between groups.

Planned alternation by arrival did not appear as a strictly alternating recorded sequence. The study is not described as randomized.

## 03 MEASUREMENT

| | |
|---|---|
| **Engagement**<br>Absorption during the visit | **Interest**<br>Interest maintained throughout |
| **Learning**<br>Learned new knowledge | **Understanding**<br>Deeper exhibit understanding |
| **Reflection**<br>Prompted further thought | **Meaning**<br>Visit felt meaningful |
| **Context**<br>Connected exhibits to background | **Emotion**<br>Emotional response to exhibits |

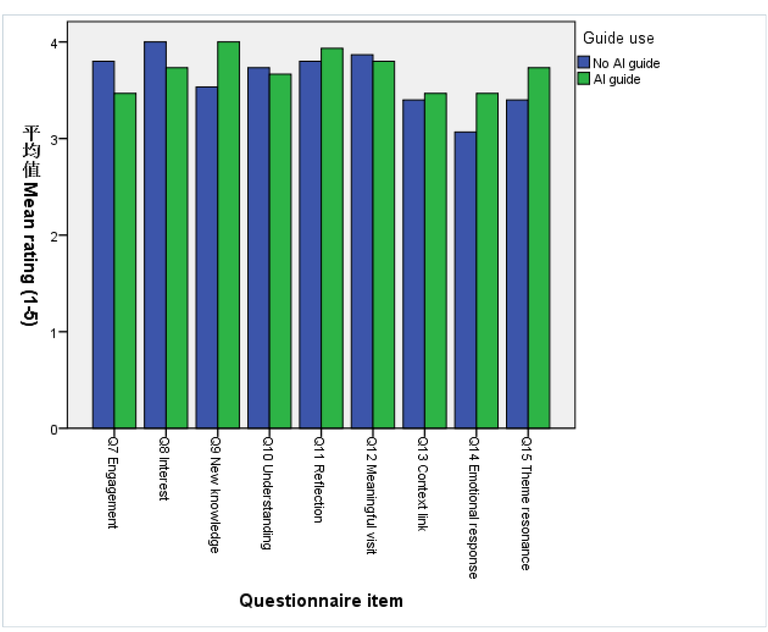
## 04 ANALYSIS

Two-sided independent-samples Mann–Whitney U tests for each ordinal item. Holm correction covers all nine comparisons. Rank-biserial correlation reports effect direction and magnitude.

| 15 / 15 | 270 / 270 |
|---|---|
| No AI / AI participants | Valid Q7–Q15 ratings |

No missing ratings, duplicate IDs, or out-of-range values were found.

## 05 RESULTS CHART




Blue: no AI guide. Green: AI guide. Bar height is descriptive; inference uses ranks.

## 06 KEY RESULT

| 5 | 4 |
|---|---|
| Items with higher AI-group means | Items with higher no-AI means |

Q9 had the largest descriptive difference: +0.47, raw p = .230, Holm p = 1.000, $r_{rb}$ = +.249.

No item was significant before or after correction. This is absence of evidence for a difference, not proof of equivalence.

## 07 FULL RESULT PROFILE

| ITEM | Δ AI−NO AI | RAW P | HOLM P |
|---|---|---|---|
| Q7 Engagement | −0.33 | .278 | 1.000 |
| Q8 Interest | −0.27 | .328 | 1.000 |
| Q9 New knowledge | +0.47 | .230 | 1.000 |
| Q10 Understanding | −0.07 | .804 | 1.000 |
| Q11 Reflection | +0.13 | .757 | 1.000 |
| Q12 Meaning | −0.07 | .622 | 1.000 |
| Q13 Context | +0.07 | .875 | 1.000 |
| Q14 Emotion | +0.40 | .318 | 1.000 |
| Q15 Resonance | +0.33 | .427 | 1.000 |

## 08 INTERPRETATION

The current data do not provide sufficient evidence that AI-guide use changes any of the nine post-visit ratings.

- All outcomes are self-report.
- The 30-student sample is small.
- No verified primary endpoint was available.
- Allocation records prevent causal interpretation.

## 09 NEXT STUDY

1. Verify the allocation process.
2. Pre-specify one primary outcome.
3. Recruit a larger sample.
4. Keep the exhibition context comparable.

## 10 CONCLUSION

**No significant difference in this sample.** Do not claim equivalence or a causal AI effect. The main methodological lesson is that allocation records, a pre-specified endpoint, and correction for multiple tests must be planned before data collection.